

The Vent-Snap Ratchet: Quantifying, Modeling, and Eliminating Seal-Rupture Slip in Suction-Based Wall Climbing Without Force Sensing

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Abstract—Legged robots that climb with vacuum suction cups must break a seal at every step. On compliant, position-controlled platforms we show that this seal rupture releases elastic energy stored in the leg chains: the body drops within tens of milliseconds and the next adhesion locks the loss in place, so that a robot stepping in place with zero net command descends by a ratchet. Using a low-cost phone-video pipeline (normalized cross-correlation tracking, per-run metric scale, command-marker time synchronization), we quantify the effect at millimeter and per-leg resolution: 93% of the per-cycle position loss is incurred at the seal ruptures, each a drop completed within two video frames (≤ 67 ms). A one-dimensional parallel-spring model explains the loss and makes four predictions that we verify, including strict linearity of the rupture bounce in an unloading displacement applied before venting. Building on this, we propose the *zero-force handover*: before a leg is vented, its foot is commanded up the wall by a per-leg calibrated amount δ_j —the deflection its leg chain has stored—while the supporting feet are commanded down the wall by the same total amount, split by weights that favor the legs farthest from their own next lift, so that the commanded body position is unchanged. The scheme is purely kinematic and open loop: the unloading amount comes from an offline table, with no force, tactile, or current feedback, and the cup-pressure sensors already used to sequence adhesion play no part in it. In an $n=3$ Latin-square study it cuts total slip by 43% (δ alone) and 61% (with weights), and slip at the rupture instants by 82%/88% (paired $p \leq 0.0022$). A pre-registered rate-discrimination experiment shows that the residual loss incurred during the handover scales with the displacement transferred, not with the time taken ($\Delta b = \kappa \delta_j$, κ reproduced to 4% across five days), identifying load-share allocation—not speed—as the remaining lever (κ reduced 44% by the weights). Data, logs, and the measurement pipeline are released.

Index Terms—Climbing robots, legged robots, compliance and impedance control, calibration and identification, mechanism design.

I. INTRODUCTION

LEGGED climbing robots that adhere by vacuum must do something wheeled and sliding-suction climbers avoid: at every single step they destroy an adhesion contact and rebuild it elsewhere. For negative-pressure feet the destruction is not gradual. A vented cup holds its grip while residual vacuum persists and then lets go in one instant when the seal lip peels—a *rupture-type* detachment that concentrates the mechanical consequences of releasing a loaded contact into a few milliseconds, far below the control bandwidth of any gait controller.

On a stiff robot this would matter little. But small wall-climbing platforms are not stiff: safety margins are bought with lightweight, 3D-printed leg chains, bellows cups that conform to the surface, and low-cost position-controlled servos with no torque feedback. On such a platform—a six-legged walker with a suction cup on each foot (Fig. 1)—we observed that when the robot is commanded to *step in place* on vertical glass—lift one foot at a time and put it back exactly where it was—it slides down the wall by 57–74 mm per gait cycle, although the attached cups never slip on the glass and the body never drifts while all six feet are attached. Frame-by-frame decomposition shows that in the $n=3$ study of this letter 93% of the loss is incurred at the seal ruptures (83% in the *discovery run*, the earlier run in which we first observed the effect, which used wider segmentation windows); each rupture drop completes within two video frames (≤ 67 ms), while during active venting, before the seal lets go, the body moves less than 0.15 mm. The mechanism is elastic: the attached feet stay put while the body sags under its own weight, so each attached leg chain—the printed links, joints, and servo gear trains between foot and body—is stretched like a loaded spring; venting a loaded leg cuts that spring at full tension, the body falls until the additional stretch of the remaining legs catches it, and the next adhesion, which grips wherever the body then hangs, locks the new position in. We call the resulting staircase descent the *vent-snap ratchet*. The current drawn by the servos corroborates the picture: the total bus current climbs step-wise from 0.7 A to ~ 1.9 A over two cycles and does not relax, indicating internal forces that keep growing between the legs.

The failure combination—compliant support chains, rupture-type detachment, and position-domain actuation that accumulates internal forces [1]—is generic to low-cost suction climbers, yet we found no prior quantification of the resulting whole-body position loss (Sec. II). Prior treatments of detachment disturbance measure foot forces or body vibration and manage them with force sensors in closed loop; the classical walking-machine literature unloads a leg before lift-off entirely in the force domain. Our platform has no force, tactile, or per-joint current sensing at all, and retrofitting it would add weight, wiring, and complexity, all of which are costly on a climbing robot.

Instead we exploit a simple constraint: an attached foot cannot slip, so *changing its position command changes force, not position*. The unloading condition “lift only a force-free leg” can therefore be translated from the force domain into the

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displacement domain and executed open loop: before venting leg j , command its foot up the wall by a per-leg constant δ_j —the stored deflection, calibrated offline by linear extrapolation of the rupture bounce—which, because the foot itself cannot move, relaxes the leg’s spring; at the same time command the supporting feet down the wall by the same total displacement, split by weights, so that the mean of the six foot commands—and with it the commanded body position—is unchanged. We call this the *zero-force handover* (Fig. 1c).

The contributions of this letter are:

1) **Discovery and quantification of the vent-snap ratchet.**

To the best of our knowledge, the first time-resolved, per-leg, millimeter quantification of whole-body position loss at the instant of vacuum-seal rupture in a legged suction climber, with a reproducible low-cost video pipeline and released data. Prior reports of detachment disturbance are qualitative for body motion, or quantify foot forces, accelerations, or cavity pressure instead [2], [3], [4].

2) **A parallel-spring account with verified predictions.** A one-dimensional elastic model explains the loss, predicts strict linearity of the rupture bounce in the unloading displacement applied before venting (verified; per-leg slopes spread $4\times$ and furnish a stiffness estimate that agrees with an independent route, $r = 0.938$), predicts the dilution effect of load-share weights (-32% predicted vs. -30 to -36% measured across metrics), and—via a pre-registered discrimination experiment—yields a transfer-loss law $\Delta b = \kappa \delta_j (1 + 0.20(r-1))$ (r the gait-cycle index) in which time plays no role.

3) **A force-sensor-free, open-loop, software-only remedy with $n=3$ evidence.** The zero-force handover converts pre-detachment unloading—known in force-sensing closed-loop forms [5], [6]—into the displacement domain: offline per-leg calibration, mean-invariant apportionment, rotation weights, and no sensor of any kind in the unloading loop (the cup pressure sensors only sequence adhesion, as in the baseline). In a Latin-square $n=3$ study on glass it reduces total per-cycle slip by $43\%/61\%$ and rupture-segment slip by $82\%/88\%$ (all pairwise comparisons $p \leq 0.0022$, surviving Bonferroni correction).

All claims of priority are made “to the best of our knowledge” and are delimited against the nearest prior work in Sec. II. Logs, videos, verification crops, and the analysis pipeline will be released **[TODO: package the open-data bundle; add repository URL]**.

II. RELATED WORK

1) *Platforms*: Adhesive legged climbers divide by attachment principle into vacuum/negative-pressure feet [7], [8], [9], [10], dry/gecko-inspired adhesion [11], [2], [12], and grasping or spine-based feet [13], [6]; see surveys [14], [15]. This letter concerns the oldest line, discrete-gait suction feet, where each step must break and re-make a seal—a cost that sliding-suction designs [16], [17] avoid at the expense of continuous pumping. Our platform is a low-cost, 3-DOF-per-leg hexapod in this lineage.

2) *Detachment disturbance: what has been measured*: That releasing an adhesive foot disturbs the rest of the robot is known qualitatively: Stickybot’s designers report that with isotropic pads the large detachment force “produc[ed] oscillations that frequently caused the other feet to slip” [2], and a recent Chinese survey names “transient and large detachment forces transmitted to the trunk causing body shudder” an open problem [3]. What has been *quantified* is foot-level: peel-angle impedance control reduces normal detachment force [18], rotational peeling cuts pull-off force by 65.5% [19], and trajectory/force shaping reduces maximum detachment force by 28% and IMU-measured vibration by 82% [20]. We note explicitly that the 82% in [20] is an acceleration amplitude (m/s^2) and numerically coincides with, but is dimensionally unrelated to, our 82% reduction of rupture-segment *displacement* (mm). In the vacuum lineage the only transient quantification we found is of cavity pressure [4]. Whole-body position loss per detachment event—the quantity that actually bounds climbing progress—appears in no prior work we could locate.

3) *Unloading before detachment*: The idea of unloading a foot before releasing it has force-domain, closed-loop precedents, against which we delimit our claim. The Stickybot patent describes tangentially unloading a foot before lift-off to relax accumulated force and elastic deformation, using Hall-effect force sensors in a stiffness-control loop [5]. LORIS’s gait state machine contains an explicit *Unload* step that constrains the departing gripper’s contact force to zero in an on-line force optimization [6]. LEMUR 3 manages attach/detach forces with a per-limb force sensor [13]; admittance control mitigates reaction forces on limbed climbers in simulation [21]; reaction-aware planning shapes swing inertia [22]; quasi-static gait optimization redistributes contact forces at the planning level [23]. All of these execute in the force domain, in closed loop, or at the planning layer, and none quantifies body slip. Our claim is accordingly narrow: converting the unloading condition to the *displacement domain* (per-leg offline calibration, mean-invariant apportionment), executing it *open loop on a platform with no force sensing whatsoever*, and validating it by *rupture-segment slip*.

4) *Force distribution in walking machines*: Smoothly driving a leg’s force to zero before lift-off is a classical walking-machine topic from Waldron’s force/motion management onward [24], [25], [26], [27], [28], through to modern whole-body controllers and MPC [29], [30], [31], [32]. The entire line executes in the force domain: either force/torque sensors close the loop [33], [1], or force setpoints are handed to actuators that can track them. Chen et al.’s “does not require force sensors” [34] refers to the setpoint computation only. The DLR Crawler paper states the problem addressed here—position-controlled multi-legged robots accumulate internal forces—and solves it with joint torque sensing [1]. Insects face the same problem: load transfer to neighboring legs mechanically unloads a leg, and load sensors trigger its swing [35]—a biological closed loop that our open-loop table replaces. In this classical frame, our contribution is a new execution domain for an old requirement, on a platform class (hobby position servos, strong series compliance) the classical solutions do not cover.

5) Displacement-domain feed-forward for compliance:

Calibrate-then-feed-forward compensation of structural compliance is itself a mature method family: TITAN XI pre-calibrates leg flexibility to restore foothold accuracy on slopes [36]; Ota et al. compensate gravity-induced deformation on a suction quadruped for motion smoothness [37]; machining robots correct commanded trajectories from offline stiffness models [38], [39]; legged humanoids feed forward sole/joint deformation [40]; a gecko-inspired climber applies stiffness-model-based gravity compensation for posture [41]; sway compensation shifts the body to transfer *gravity* load in static gaits [42]. Same mathematical shape, different target: none of these uses calibrated displacement to *zero the internal force of a leg about to detach*, and none faces energy release concentrated into milliseconds.

6) *Alternative routes*: One can avoid rupture altogether (sliding suction [17], at the cost of continuous power), estimate contact force from motor current [43] (infeasible on hobby servos with large gearbox friction; see Sec. VII), or exploit snap-through energy release for propulsion rather than as a disturbance [44]. Directional dry adhesives detach at near-zero force by material design [2], [45]; vacuum cups have no such mechanism, so the stored energy must be *returned before rupture*. Slow venting, the intuitive remedy, does not help on our data (Sec. III-C).

III. PLATFORM, PIPELINE, AND THE PHENOMENON

A. Platform

The testbed (Fig. 1) is a six-legged 3D-printed walker derived from the open-source MakeYourPet hexapod kit [46] (printed body and coxa/femur links), fitted with self-designed printed tibia-suction-cup legs and a pneumatics bay. The robot climbs facing up the wall; legs are named L/R for the left/right side and 1/2/3 from front to rear, so L1 and R1 are the uppermost pair and L3/R3 the lowest (Fig. 1a). Each leg has three position-controlled joints (coxa/femur/tibia links 43/81/124 mm, as measured and used by the kinematics) and ends in a 30 mm bellows suction cup. A normally-closed 3-way solenoid valve per foot connects the cup either to the pump manifold (attach; the de-energized, fail-safe state) or to atmosphere (vent), and a per-foot check valve between manifold and valve isolates the cups from one another. No vacuum reservoir was fitted in any run reported here: a single diaphragm pump works directly on the manifold and holds the shallowest attached cup between -60 and -75 kPa by bang-bang hysteresis on the cup-pressure readings (and runs unconditionally while any cup is being evacuated); a cup counts as attached below -30 kPa (held for 0.2 s); a leg may lift only when the five other cups are attached and deeper than a -50 kPa gate. The gait moves one leg at a time: at any moment five feet are attached and at most one is in the air. Eighteen 35 kg-cm position-controlled hobby servos (no torque, force, tactile, or per-joint current feedback of any kind) are driven by a Raspberry Pi 5 through an RP2040 servo board running open-source firmware [47]; total mass is 3.54 kg. The robot's sensors are six cup-pressure transducers (one per foot), which the adhesion state machine reads for

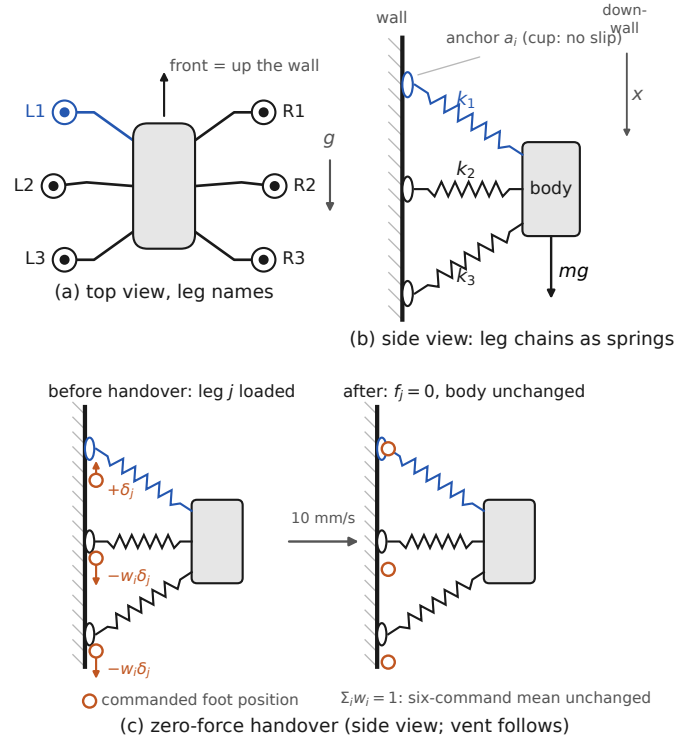


Fig. 1. Platform and the zero-force handover (schematic). (a) Top view and leg naming: the robot faces up the wall, so legs 1 are the top pair. (b) Side view: each attached leg chain acts as a spring k_i between the body and a foot anchor a_i that cannot slip; the body hangs on the springs under its weight mg ; x is measured down the wall. (c) Zero-force handover, shown for a top leg j : every attached foot sits above its commanded position (open circles), i.e. every leg is stretched. Before venting j , its foot command moves up the wall by δ_j until the spring is relaxed ($f_j = 0$), while the supporting feet's commands move down the wall by $w_i \delta_j$, $\sum_i w_i = 1$; the commanded body position is unchanged and the seal then breaks with nothing to release. [TODO: add (d): photograph of the robot on the glass wall.]

attachment and vent confirmation, leak detection, the lift gate, and pump hysteresis, and the servo board's bus voltage/current telemetry, used for under-voltage shutdown and logging only; there is no IMU, no foot contact switch in use, and no servo position read-back. None of these readings enters the handover computation (Sec. V). All experiments run on vertical glass with a safety rope. The platform's compliance is inherent to its construction: printed legs, bellows cups, and servo gear trains together make the chain from body to foot anchor soft enough that commanded body motion is only 42% realized (Sec. VI-D)—typical of the lightweight construction this robot class requires.

B. Measurement pipeline

We measure body position along the wall from a fixed phone camera (4K@30 fps) by normalized cross-correlation (NCC) template tracking of the robot's electronics board, with sub-pixel parabolic interpolation and frame-wise compensation of camera motion against a static reference patch. Metric scale is taken *per run* from the long edge (93.0 mm) of a PCB rigidly mounted on the robot, localized by four-corner sub-pixel fits (0.288–0.296 mm/px across runs; the top and bottom

edge lengths differ by 1.1–2.7% and the scale is taken as their mean). Video time is registered to the robot’s on-board event log by a commanded time-sync marker, the *body lean*: with all six feet attached, the body is commanded 10 mm up the wall and back at the start of each run, and the commanded profile is fit in shape to the tracked trajectory (R^2 0.95–1.00 per run); the same marker doubles as a compliance probe (Sec. VI-D). Every lift event is then decomposed into five consecutive time segments: *handover* (the pre-vent displacement transfer of Sec. V, when enabled), *vent* (from 1 s before the valve command in the baseline, or from the valve command itself in the handover conditions, where it closes the handover segment, to 0.4 s after the logged release confirmation—cup pressure back above -5 kPa, “pressure-zero” below; the window contains the seal rupture, and we call the slip accrued in it the *rupture-segment slip*), *swing* (foot in the air), *press* (touchdown), and *settle*. Segment sums were checked against manual tape-ruler readings (discovery run: 131.3 mm tracked vs. 131 mm read), and all scale boards and template locks were human-checked against 188 archived localization crops.

C. The vent-snap ratchet

Stepping in place isolates the effect: each leg in turn, in a fixed *rotation* order (R3→L1→R2→L3→R1→L2; one pass through all six legs is a *round*, i.e. one gait cycle), is raised to a commanded 15 mm above the glass plane (33 mm of foot-command travel, since the attached command sits 18 mm into the surface as press pre-load), hovers 1 s, and is returned to its nominal station; the commanded net body motion is zero, so any systematic displacement must come from elastic strain release and re-locking. Two controls verify the premises: operator-checked marks on the glass beside each attached cup do not move (the cups do not slip), and the body does not creep while all six feet are attached (the settle segments contribute only 2.4–4.7 mm per three-round run; see also Sec. VI-E). Yet the body descends 57–74 mm per cycle (discovery run; 63.9 ± 0.9 mm in the $n=3$ baseline, Sec. VI-B), in a staircase whose every step coincides with a seal rupture (Fig. 2).

Three time scales matter. (i) While residual vacuum drains after the venting command, the body does not move (< 0.15 mm). (ii) At seal rupture the entire drop completes within 1–2 frames (33–67 ms), with an underdamped overshoot (Fig. 2b: peak 26.7 mm, settling at 21.5 mm)—the signature of an elastic release. (iii) Between events the trajectory is flat. In the $n=3$ baseline the vent segments sum to 172.9 ± 3.4 of 185.5 ± 3.0 mm total—93%; touchdown press contributes 4% (discovery-run decomposition: 83%/4% with wider re-equilibration windows). The released foot itself recoils 9–15 mm down-wall at the same instant (a lower bound: the tracked patch includes part of the leg). Per-leg single-event bounces reach 20.9 mm (L1) and 19.2 mm (R1)—the top pair, where the body’s weight exerts the largest peel moment on the cups—versus 3.5–6.1 mm for the middle pair (L2/R2), the softest legs, a first sign that leg-chain stiffness is strongly unequal.

Two corollaries constrain the design space. First, *slow venting cannot help*: the data show no motion while residual

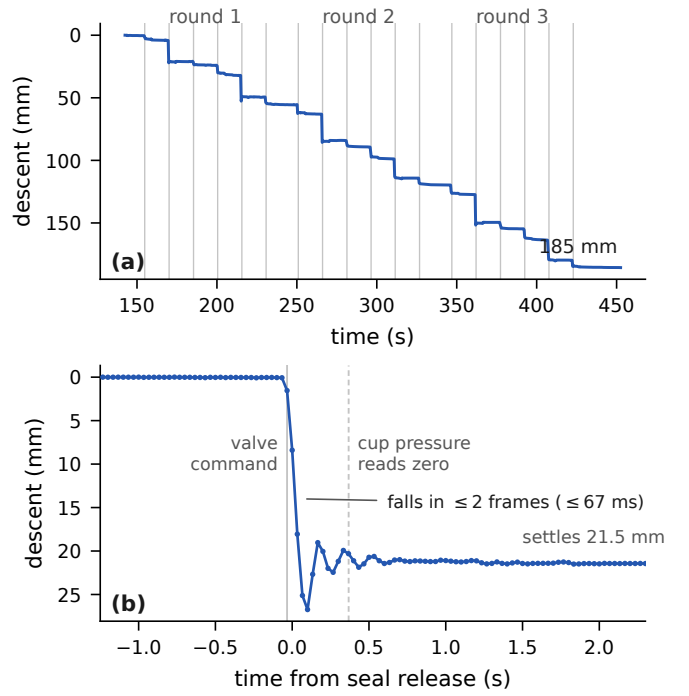


Fig. 2. The vent-snap ratchet (baseline, replication #1). (a) Body position along the wall over three rounds of stepping in place: every step of the 185 mm staircase coincides with a seal rupture (gray lines); the trace is flat between events. (b) Native-30 fps profile of one lift (L1, round 2), $t=0$ at the tracked release: the body is stationary before the release, falls 21.5 mm within two frames with an underdamped overshoot (peak 26.7 mm); the released foot simultaneously recoils 12.8 mm down-wall (not shown). The logged valve command and release-confirmation (pressure-zero) events bracket the mechanical release; their placement relative to the video carries the sync uncertainty (a few tenths of a second).

vacuum drains, so the force is not released gradually as the pressure decays but all at once when the lip peels; venting more slowly would only postpone the same step. (We did not test slower venting directly; the inference rests on the flat pre-rupture trace and the no-slip control.) Any remedy must therefore ensure the leg carries no force *at the moment the seal lets go*. Second, the loss is locked in by the next adhesion: a fresh cup grips wherever the body then hangs, at zero preload (Sec. IV), so the drops accumulate as a ratchet rather than averaging out. Servo bus current shows the same accumulation: from 0.73 A at first attachment it climbs step-wise to ~ 1.9 A after two cycles and does not relax—each locked-in drop leaves the support legs fighting a larger internal force.

IV. A PARALLEL-SPRING ACCOUNT

A. Model

Collapse each attached leg chain i to a linear spring of stiffness k_i along the wall’s gravity axis (x positive down; Fig. 1b). Let a_i be the (immovable) anchor of foot i and c_i the foot position the controller commands, expressed relative to the body, and define $y_i \equiv a_i - c_i$: the body position at which the foot would sit exactly where commanded, so that leg i carries zero force. Leg i then exerts the shear force

$f_i = k_i(x - y_i)$ on the body (positive up-wall), and static equilibrium of the attached set S gives

$$x = \langle y \rangle_k + \frac{mg}{K}, \quad \langle y \rangle_k \equiv \frac{\sum_{i \in S} k_i y_i}{K}, \quad K \equiv \sum_{i \in S} k_i. \quad (1)$$

The body hangs below the *stiffness-weighted* mean of the legs' zero-force positions. Three events drive the dynamics:

Zero-preload lock-in. A cup that seals grips wherever the body currently is, storing no force: adhesion sets $y_j \leftarrow x$.

Rupture. Venting leg j removes a spring carrying tension f_j ; the body falls to the new equilibrium of the catching set,

$$b_j = \frac{f_j}{K - k_j}, \quad (2)$$

i.e. *the observed bounce is the leg's stored force divided by the parallel stiffness of the other legs*—not the leg's own stored deflection $e_j = f_j/k_j$. The ratio $e_j/b_j = (K - k_j)/k_j$ ($= 5$ for six equal legs) ranges from 1.7 for the stiffest leg to 8 for the softest on our platform (Sec. V-B), a distinction that matters for calibration.

Loading. Between adhesion and lift, every millimeter the body descends stretches each attached leg by one millimeter ($x - y_i$ grows); legs attached longer ago are stretched more, so the legs' stored forces differ even under identical commands.

Iterating lock-in, loading, and rupture over the gait in a one-dimensional simulation reproduces the ratchet: the steady-state in-place loss is $\approx 0.4 mg/\bar{k}$ per cycle, and matching the observed 74 mm/cycle gives a whole-robot compliance scale of $mg/\bar{k} \approx 185$ mm. Stiff-leg bounces plateau at ~ 20 – 27 mm and do not grow past round 2, and the linear model at $mg/\bar{k} = 185$ mm over-predicts the loss in forward climbing (it predicts net retreat, whereas the robot nets +26% of commanded stride); we attribute both to servo torque saturation capping the energy a single leg can store—a hypothesis, not a measurement [TODO: net-advance demo (T4b) will refresh this number].

B. Four verified predictions

P1: The rupture bounce is strictly linear in an unloading displacement applied before venting, with slope encoding leg stiffness. If, before venting, the foot of leg j is commanded δ up the wall, moving y_j toward x by δ (Sec. V), the residual tension falls by $k_j\delta$, so $b_j(\delta) = b_j(0) - s_j\delta$ with $s_j = k_j/(K - k_j)$. A three-dose calibration experiment ($\delta = 0/12/24$ mm, the same for all six legs; Sec. V-B, Fig. 3) finds all six legs strictly linear (mid-dose residual ≤ 1 mm), with slopes 0.13–0.57—a $4\times$ per-leg spread that directly measures the stiffness imbalance. The normalized stiffnesses $\hat{k}_j = s_j/(1 + s_j)$ from these slopes agree with a completely independent second route (fitting handover-segment sag, Sec. VI-E) at Pearson $r = 0.938$.

P2: A freshly attached leg bounces by little. Zero-preload lock-in implies the first lift of a just-attached leg releases only the force accrued since lock-in: the first lift of every baseline run (R3) shows that leg's smallest bounce—0.0 and +1.8 mm in the two runs without a pre-run marker, 2.5–4.3 mm in the three main-study baselines, where a ± 10 mm

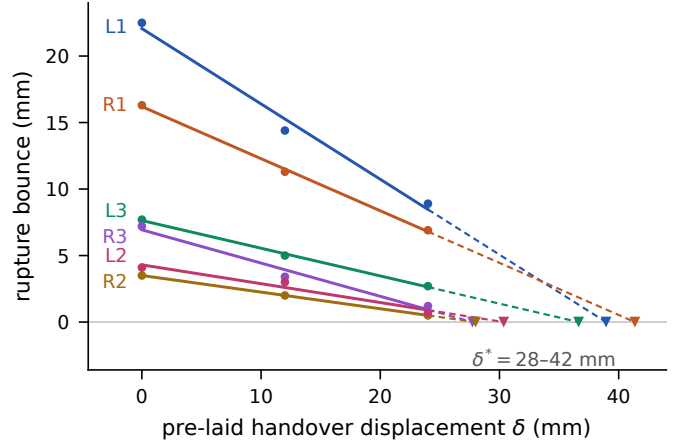


Fig. 3. P1 verified: rupture bounce vs. unloading displacement δ applied before venting (round 2 of the three-dose calibration, $\delta = 0/12/24$ mm, the same for all legs). All six legs are strictly linear (mid-dose residual ≤ 1 mm); the slopes spread $4\times$ across legs (stiffness imbalance), and the dashed extrapolations to zero bounce define the calibration targets $\delta_j^* = 28$ – 42 mm (triangles).

marker precedes it—against 6–8 mm in later rounds as loading accumulates.

P3: Equilibrium is stiffness-weighted, so mean-invariant commands do not perfectly freeze the body. By (1), a handover that moves y_j up by δ and the supports down by weights w_i ($\sum w_i = 1$) shifts equilibrium by

$$\Delta x = \frac{\delta(k_j - \sum_{i \in S \setminus j} k_i w_i)}{K}, \quad (3)$$

zero only for equal stiffness. Unloading the *stiffest* leg must sink the body; the first-order prediction for L1 (P1 slope stiffnesses, uniform shares) is 4.6 mm vs. 6.8 mm measured (first round), and the fitted form of (3), once the common-mode term of (4) is included, reproduces all 54 leg $\times$ round $\times$ replication cells of the B condition with RMSE 0.85 mm (Sec. VI-E).

P4: Load-share weighting dilutes stored energy. Because each lock-in re-zeros one leg, load handed to a support early is partially re-absorbed by subsequent lock-ins, while load handed late is carried intact into that leg's own lift. Weighting the handover shares toward the legs *farthest from their next lift* (most dilution opportunities ahead) lowers steady-state stored energy: the 1-D model gives -19% for a mild schedule (0.35/0.25/0.2/0.15/0.05), -32% for the deployed schedule (0.6/0.25/0.1/0.05/0), and $+30\%$ for the mild schedule reversed (worse than uniform). The measured B $\rightarrow$ C reduction of residual slip is -30 to -36% across metrics (-28 to -33% per replication on totals; Sec. VI-B).

V. ZERO-FORCE HANDOVER

A. Displacement-domain unloading

The remedy follows from one invariant: an attached cup does not slip (verified, Sec. III-C), so foot commands move forces, not feet. Before venting leg j , the gait controller inserts a HANDOVER phase (Fig. 1c): the lifting leg's foot command moves *up the wall* by δ_j (returning its stored deflection), while each supporting foot command moves *down the wall*

by $w_i \delta_j$ with $\sum_i w_i = 1$ (taking over the load); both are applied quasi-statically at 10 mm/s along the wall's vertical axis (the robot's forward axis, since it faces up the wall). The mean of the six foot commands is unchanged, so the body is commanded to stay put; the lifting leg's force is driven to zero; the seal then ruptures with nothing to release. The supports never carry more than they would carry *after* the lift anyway—the handover only moves the transfer earlier and makes it lossless. Commands remain bounded: over one gait cycle each leg receives shares totalling roughly one δ down the wall and gives back its own δ_j when its turn comes, and the per-leg mismatch is cleared when the foot is returned to its nominal station at touchdown. Simulating the command stream with the deployed tables, the net effect is a one-time offset of the six-command mean accrued during round 1 (≈ 14 mm up the wall with uniform shares, ≈ 20 mm with rotation weights, i.e. ≈ 6 – 9 mm of actual body motion at the measured transfer ratio), after which the mean oscillates within ± 0.9 mm with zero net drift per round—one reason round 1 is excluded from judgment (Sec. VI-A).

B. Per-leg calibration: bounce is not stored energy

The naive calibration—set δ_j to the observed bounce—fails by the factor in (2): the bounce is the stored force read through the *other* legs' parallel stiffness, and underestimates the stored deflection by 1.7– $8\times$ depending on the leg. The correct, scale-free procedure follows from P1: run three doses ($\delta = 0$, half, full), fit the per-leg line, and extrapolate to the zero-bounce intercept $\delta_j^* = b_j(0)/s_j$. Measured $\delta^* = 28$ – 42 mm per leg; we deploy $\delta_j = 0.8 \delta_j^*$, deliberately under-shooting (a small residual bounce is benign, whereas overshooting would pre-load the leg in the opposite direction), giving the table L1/R1/L3/R3/R2/L2 = 31/33/29/22/22/24 mm. Because δ^* is a ratio of same-unit quantities, it is independent of the video scale calibration—the least scale-sensitive quantity reported here. The response to dose is monotone ($65.6 \rightarrow 50.0 \rightarrow 38.7$ mm per cycle across 0/12/24), and the staircase character is unchanged: δ lowers each step's height, not its timing.

C. Rotation-distance load-share weights

P4 fixes the apportionment: shares are assigned by *rotation distance*, i.e. by how many lifts away each support is from its own next lift—the support that lifted most recently (farthest from its next lift) receives the largest share and the next leg due to lift receives none (0.6/0.25/0.1/0.05/0, listed from most recently lifted to next-to-lift). The controller re-normalizes the shares over the current support set at every control tick and falls back to uniform shares (with a logged note) if the lift order ever deviates from the rotation—the weights assume the cyclic gait, and misapplying them would concentrate load on one leg. Weights change the steady-state operating point; in practice the table calibrated without weights needed no correction under them, and the same table is used in conditions B and C (Sec. VI-A).

D. Implementation and guards

The method is ≈ 150 lines of controller code (≈ 270 with comments): one new phase between the lift decision and

TABLE I
MAIN RESULTS, $n=3$ LATIN SQUARE (MM; MEAN $\pm$ SAMPLE SD OVER REPLICATIONS). PERCENTAGES VS. A. A ROUND IS ONE PASS THROUGH ALL SIX LEGS; RUPTURE SEGMENT = SLIP ACCRUED IN THE VENT WINDOWS; HANDOVER SEGMENT = SLIP DURING THE PRE-VENT TRANSFER; CURRENT RISE = SERVO BUS CURRENT AT THE END OF A RUN MINUS AT ITS START.

	A (baseline)	B (+ δ)	C (+weights)
Round-2 slip	63.9 $\pm$ 0.9	38.1 $\pm$ 1.0	26.5 $\pm$ 1.7
3-round total	185.5 $\pm$ 3.0	105.6 $\pm$ 2.2 (−43%)	72.9 $\pm$ 3.0 (−61%)
Rupture segment	172.9 $\pm$ 3.4	31.6 $\pm$ 1.0 (−82%)	20.1 $\pm$ 3.3 (−88%)
Handover segment	—	60.2 $\pm$ 2.2	39.7 $\pm$ 4.3
Current rise (A)	+0.97 $\pm$ 0.02	+2.40 $\pm$ 0.08	+1.85 $\pm$ 0.08

VENT, one per-tick displacement update, no new hardware, no added sensing, no per-robot tuning beyond the δ table. Safety guards (none of which fired in any run reported here): the displacement transfer pauses while the automatic leak-rescue routine re-pumps a cup whose vacuum is decaying; every per-tick foot target is checked against the leg's workspace envelope, and the handover is truncated (partial unloading, logged) rather than driving a support out of its workspace; and the -50 kPa lift gate is re-checked after the transfer, since seconds have passed since the lift decision. The handover adds $\delta_j/(10 \text{ mm/s}) \approx 2.2$ – 3.3 s per step at the deployed table—the cost in cycle time; the energy cost is quantified in Sec. VI-F.

VI. EXPERIMENTS

A. Protocol

Conditions. A: baseline stepping in place; B: A + zero-force handover with the calibrated table ($\delta_j = 0.8 \delta_j^*$) and uniform shares ($w_i = 1/5$); C: B + rotation-distance weights (same δ table). One variable changes per condition.

Design. $n=3$ full-protocol replications, each an independent mount–run–dismount of all three conditions, ordered by a 3×3 Latin square (#1 A $\rightarrow$ B $\rightarrow$ C, #2 B $\rightarrow$ C $\rightarrow$ A, #3 C $\rightarrow$ A $\rightarrow$ B) so that condition is decoupled from within-session order, battery state, and glass condition. Each run: after full attachment the robot rested 21–58 s (operator-timed; a 60 s settle was built into the launch sequence only after these runs and applies to the D' series of Sec. VI-E); one keypress then launches a fixed automated sequence— ± 10 mm time-sync marker, then 3 rounds $\times$ 6 legs of lift–hover–replace paced at ≥ 10 s between legs and only once the pump had been off for ≥ 2 s (the 10 s floor governed every interval), 15 s between rounds, 30 s final settle—so pacing is identical across runs by construction. Battery ≥ 7.88 V at each session start; camera untouched within a run (it was moved between runs in two sessions, Sec. VI-G); scale re-measured per run. All nine logs show no safety freezes or leak-rescue pauses, no envelope truncations, and no lift-gate waits. Judgment uses round 2 by protocol (round 1 contains the known ramp-up and command-reset artifacts; round 3 is the replicate check).

B. Main results

Table I and Fig. 4 summarize the main result. Paired per-replication differences (df = 2, two-sided): B–A = $-79.9 \pm$

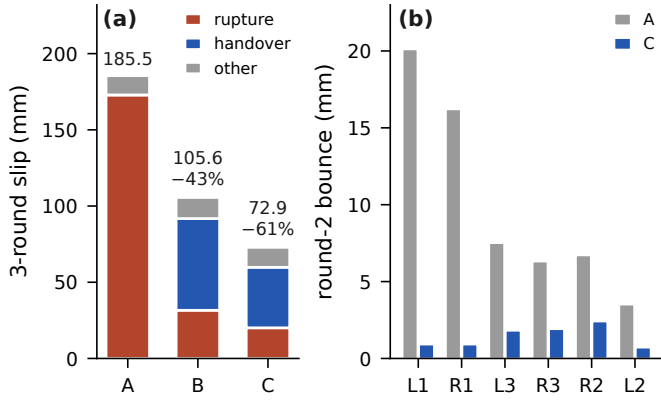


Fig. 4. Distribution of the loss ($n=3$ means). (a) Three-round slip by segment and condition: the handover converts an impulsive 173 mm rupture loss into a 60 mm quasi-static transfer loss; weights compress it to 40 mm. (b) Per-leg round-2 rupture bounce, A vs. C: the stiff front legs drop from 20.1/16.2 to 0.9/0.9 mm.

5.0 mm ($t = 27.9$, $p = 0.0013$); $C-A = -112.6 \pm 5.9$ mm ($t = 32.9$, $p = 0.00092$); $C-B = -32.7 \pm 2.7$ mm ($t = 21.4$, $p = 0.0022$). All three survive Bonferroni correction for the three comparisons ($p \times 3 < 0.007$). We emphasize what $n=3$ means here: three *full-protocol* independent replications, each containing 18 lift events per condition, with relative effects nearly constant across replications (B/A: $-42/-45/-42\%$; C/A: $-61/-63/-58\%$)—the treatment effect is an order of magnitude larger than every noise scale in the experiment. No order effect is visible after the Latin-square decoupling.

Per-leg rupture bounces (round 2, $n=3$): the baseline reproduces the discovery run five days earlier (A-condition L1 20.1 ± 1.3 vs. 20.9); under C all legs are ≤ 2.4 mm, and the two stiff top legs, which dominated the baseline, drop to 0.9 ± 0.5 (L1) and 0.9 ± 0.3 mm (R1)—a 95% per-leg reduction on the legs that contributed most. R1 showed two slight upward bounces (≈ -0.9 mm, condition B), the signature of mild overshoot, accepted under the under-shoot rule.

C. Finding 1: the residual is in the transfer, on the stiffest legs, and grows

With rupture slip suppressed, roughly half to three-fifths of the B/C residual (B 56–58%, C 47–62% per replication) sits in the *handover segments*, concentrated on the two stiffest legs and, for L1, growing round over round (B-condition L1: $6.8 \rightarrow 10.3 \rightarrow 12.6$, $6.9 \rightarrow 8.6 \rightarrow 10.6$, $6.7 \rightarrow 8.8 \rightarrow 11.9$ mm across the three replications; R1 is of the same size, $6.6-11.4$ mm, but not monotone; the other four legs 0–2 mm). The first-order reading—P3’s stiffness-weighted equilibrium—is correct about the *shape* but, as the modeling below shows, wrong about the *size*: redistribution explains only ~ 7 mm of the 60 mm segment. The decomposition of this residual, and which lever actually moves it, is the subject of Sec. VI-E.

D. Finding 2: a 42% command-to-body transfer ratio

The ± 10 mm time-sync marker doubles as a compliance probe: across 9 markers in 3 sessions on 2 days and 5

camera placements, the commanded 10 mm body lean produces 4.22 ± 0.30 mm of actual body motion—a *transfer ratio* of $42.2 \pm 3.0\%$, invariant across conditions (A/B/C 4.15/4.28/4.22 mm), as expected for a probe executed at full static support. Sixty percent of commanded displacement is absorbed by the leg chains. We report this as a platform characterization, not as a new finding: it quantifies the compliance that makes the ratchet large, and it feeds the model’s overall gain (the fitted c_0 below).

E. Mechanism: fits, and a pre-registered discrimination experiment

1) *Fitting the handover segment*: Per-leg, per-round handover-segment displacements (B: 54 cells over 3 replications) are fit by

$$\Delta b_{j,r} = c_0 \delta_j (\hat{k}_j - \sum_i \hat{k}_i s_i) (1 + \gamma(r-1)) + \Delta b_{j,r}^{\text{cm}}, \quad (4)$$

the first term being (3) with a round-growth factor γ and the second a *common-mode* term shared by all legs. Within B, the model achieves RMSE 0.85 mm on signals of 6.8–12.6 mm with leave-one-replication-out CV of 0.99 mm, and the fitted $\hat{k}$ agrees with the slope route of P1 at $r = 0.938$ (final joint estimate at $c_0 = 1.15$: L1 0.265, R1 0.227, L3 0.137, R2 0.133, R3 0.125, L2 0.112; $\gamma = 0.11$; since (4) identifies only the products $c_0(\hat{k}_j - \sum_i \hat{k}_i s_i)$, the absolute spread of the $\hat{k}$ trades off against c_0 and only their ordering and ratios are robust). The resulting decomposition reverses the first-order reading: B’s 60.2 mm segment = ~ 6.7 redistribution + ~ 55.5 common-mode; C’s 39.7 = $7.7 + 30.0$ (the final four-condition fit below gives $5.7 + 54.5$ and $6.5 + 33.0$). *Most of the residual—92% in B, 76–84% in C—is a common-mode descent incurred per handover*—and the apparent concentration on stiff legs is a shape effect: on soft legs the redistribution term of (3) is *negative* (unloading a soft leg lifts the body slightly) and cancels the common-mode base, while on stiff legs the two add.

2) *What is the common mode?* A pre-registered discrimination: Two hypotheses fit the $n=3$ data equally well, because transfer time and transferred displacement are perfectly collinear at a fixed rate ($T \approx 0.5 + \delta_j/10$): (H1) *time creep* ($\Delta b^{\text{cm}} = \rho T$, loss $\propto$ window duration) vs. (H2) *per-millimeter transfer loss* ($\Delta b^{\text{cm}} = \kappa \delta_j$). Model selection on the observational data alone cannot separate them: on the $n=3$ data the two are indistinguishable ($\Delta \text{AICc} \approx 2$, marginally favouring H2), and our original model race, which lacked an H2 candidate, had selected time creep. The only discriminator is intervention: change the transfer *rate*. We pre-registered the experiment in the lab protocol with interpretation tables for both outcomes before touching the wall: three same-day C-vs-D’ pairs in alternating order, D’ identical to C except that the handover displacement is applied at 20 mm/s instead of 10 mm/s (measured handover window $3.18 \rightarrow 1.83$ s), predictions H1: handover segment $39.7 \rightarrow \sim 26$ mm (i.e. a ~ 12 mm/run improvement at no cost); H2: unchanged.

The outcome (Table II, Fig. 5) is unambiguous: doubling the rate leaves the handover segment at 36.6 ± 1.6 mm (paired D’–C = -2.9 ± 2.1 mm, $p = 0.14$; totals -1.7 ± 0.7 mm,

TABLE II
RATE DISCRIMINATION (D'): PREDICTIONS VS. OUTCOME
(HANDOVER-SEGMENT SLIP PER RUN, MM), AND JOINT MODEL
SELECTION OVER FOUR CONDITIONS (B, C, D', AND C₃₁, THE THREE
CONTROL RUNS OF THE DISCRIMINATION DAY; 216 LEG×ROUND CELLS).

	prediction	measured
H1 time creep (ρT)	26.0	
H2 per-mm ($\kappa \delta$)	≈ 39.5	36.6 ± 1.6
Joint fit (ΔAICc ; lower is better): H2 $\kappa \delta$: 0.0 ; H1 ρT : 17.8; per-condition constants: 31.6; current-based: 70.6/73.0.		

−3%), far from H1's 26.0; the outcome lands only 0.21 of the way from the H2 prediction toward the H1 prediction, and the merged handover+vent comparison gives the same non-significant picture (-3.0 ± 3.1 mm); in one pair the decrease reappeared in the vent window (a ~ 1 mm relaxation tail per event crossing the boundary at the faster rate). Aligning every handover on its vent instant (Fig. 5) shows the mechanism directly: the D' ramp is twice as steep and the *plateau is identical* (L1 7.9–8.8 mm across all six runs)—the descent tracks the millimeters transferred quasi-instantaneously and does not accumulate with window time. Sanity checks confirm that the faster rate changed nothing else: per-leg rupture bounces, transfer ratio (4.30/4.42 mm), and current increments are unchanged between conditions.

3) *The transfer-loss law and its lever*: The joint four-condition fit (H2) settles the common mode as

$$\Delta b_{j,r}^{\text{cm}} = \kappa \delta_j (1 + 0.20(r-1)), \quad (5)$$

with κ a property of the *load-share allocation*, not of time: uniform shares $\kappa_B = 0.099$ mm/mm; rotation weights $\kappa_C = 0.055$ (same-condition re-measurement five days later: 0.053—a 4% reproduction). The weights, introduced to dilute stored energy (P4), turn out to act mainly by *compressing* κ by 44%. Mechanistically the law points to a quasi-static per-millimeter loss at the cup/leg-chain interface during load transfer (micro-slip or structural nonlinearity)—categorically not time creep, consistent with the zero-drift dwell control. The practical reading: the speed of the displacement transfer does not affect the loss; share allocation is the remaining lever; and the residual handover segment ~ 37 –40 mm is the intrinsic cost of transferring $\sum_j \delta_j = 161$ mm per round at the current κ ($\kappa \sum \delta \times 3 \approx 26$ mm + redistribution ~ 6 –8 + round growth).

F. Cost

The handover converts an impulsive loss into sustained servo load: current increase over a run rises from $+0.97 \pm 0.02$ A (A) to $+2.40 \pm 0.08$ A (B), which the weights reduce to $+1.85 \pm 0.08$ A (C, −23% vs. B—direction and roughly two-thirds of the modeled −32% stored-energy dilution). Cycle time grows by the transfer duration (2.2–3.3 s per step at 10 mm/s). Both costs are the cost of adding no sensing; Sec. VII discusses the trade-off.

G. Deviations and threats to validity

We record all deviations. (i) One run's video started late, cropping the time-sync marker; the initial fallback synchro-

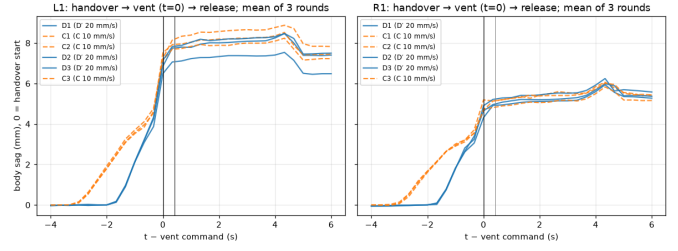


Fig. 5. The discrimination result, seen directly: handover profiles of all six runs aligned on the vent command ($t=0$), per-leg means over rounds (left L1, right R1). At double transfer rate (D', solid) the descent ramp is twice as steep as the control (C, dashed) but ends at the *same* plateau—the descent tracks the millimeters transferred, not elapsed time. **[TODO: restyle to match the other figures (fonts/colors); currently the analysis-report rendering].**

nization (regression on the periodic step pattern) locked one period off and mis-attributed per-leg slip; the error was caught by physical implausibility of the per-leg pattern, re-synchronized by a crop-tolerant marker fit ($R^2 = 0.99$, amplitude independently matching the other eight markers), and the corrected run agrees with all cross-checks. Lesson embedded in the pipeline: verify the step periodicity before using step-based synchronization; the 60 s pre-run settle is now built into the launch sequence. (ii) The camera was moved between runs in two of the three sessions (#1 A→B by ≈ 128 px, #2 C→A by tens of px, never within a run; metric scale is per-run by design, and in-run drift is compensated by the static reference (≤ 4.8 px)). (iii) Battery was not recharged between runs within a session (7.5–8.1 V range); the Latin square decouples it from condition. (iv) NCC template score medians (0.39–0.76 in the main study, 0.70–0.86 in the D' series) fall below our internal 0.8 bar in tilted-board runs; the reference-patch score (≥ 0.988) and nine-run cross-consistency (CV 1.6–4%) bound the impact. (v) Scale is referenced to the robot-mounted board plane; top/bottom-edge disagreement (1.1–2.7%) bounds the residual perspective error. (vi) In two D'-series runs a ~ 1 mm relaxation tail per event crosses the handover/vent boundary, shifting ≈ 4 –5 mm per run between the two adjacent columns; merged segments and totals are unaffected. None of these approaches the order of the effects (43–88%).

VII. DISCUSSION

Is the compliance a design flaw? The compliance is not a construction defect that could simply be designed out: lightweight printed legs, bellows cups, and unsensed gear trains are standard in few-kilogram climbers, and every element of the failure combination—compliant support chains, rupture-type detachment, position-domain actuation—recurs across the platform class (Sec. II). What the analysis adds beyond the fix is a design rule: compensability is bounded by the support set's stiffness profile. No choice of weights can zero the redistribution term (3) for the stiffest leg (with the entire share on R1, L1's coefficient is still 0.038 in normalized units, ≈ 1.4 mm per L1 handover at $\delta = 31$ mm)—so a designer who wants software-only detachment management should bound the *spread* of leg stiffnesses, not just their level. Conversely, because the remedy is a ≈ 150 -line, zero-hardware retrofit, it

is deployable on existing position-controlled climbers without modification.

Why open loop rather than current feedback? The choice is based on the data. Whole-robot current cannot be attributed per leg: during a stiff-leg handover the total shows a plateau whose onset is the balance point of unload/load gains, not the zero-force point (the leg still bounced 6.9 mm at a dose at which the current appeared to have converged); during a soft-leg handover the total rises monotonically—there is no usable feedback signal. Per-leg force sensing is exactly the hardware we set out to avoid; motor-current force estimation [43] needs friction models hobby gear trains do not honor. Meanwhile the open-loop table is estimation-free, tuning-free, and—as the $n=3$ spread shows (relative effects within 5 points across replications)—stable across days without re-calibration. The natural closed-loop upgrade is event-based, not continuous: the current step at the rupture instant (and ultimately an IMU pulse) is proportional to residual stored energy and would let the robot re-trim its own δ table between climbs; we deliberately leave this as future work so that the present result stands on no added sensing.

Model-gated experiment selection. The fitted model ruled out one of our planned conditions: stiffness-aware shares (choosing w_i to zero (3) per leg) move only the redistribution term, worth 3–4 mm, too little to justify a wall experiment. It also predicted the outcome of the rate intervention, with both interpretations fixed in advance. Requiring the model to state its prediction before each experiment proved as useful as any single result.

Generalization and limits. The phenomenon needs three ingredients: series compliance between anchor and body, detachment that releases the contact faster than the support can re-equilibrate, and actuation that tracks position, not force. Vacuum cups on any smooth surface qualify; directional dry adhesion largely does not (peel-by-design releases at near-zero force [45], [2]), which is why the problem does not arise for gecko-inspired climbers. Our evidence is one platform, one surface, an in-place proxy task, three rounds, and $n=3$: the effects are large and internally consistent, but absolute stiffness (a hang-weight calibration), net-advance climbing with the handover engaged, longer runs (the round-growth factors γ in (4) and 0.20 in (5) have not saturated at $r=3$), and the micro-mechanism of κ (whether it depends on how concentrated the shares are or on how recently the receiving cup attached, separable by a share-perturbation experiment) remain open [TODO: fold in T4a/T4b results when available]. The δ table is an open-loop calibration: payload or posture changes outside the calibrated regime require re-running the three-dose procedure (one session) or the event-based re-trim above.

VIII. CONCLUSION

Every discrete-gait suction climber loses position at the moment its seals break. On a compliant, position-controlled hexapod we measured this loss—93% of a 64 mm/cycle in-place descent concentrated into post-rupture milliseconds—traced it to elastic energy stored by the robot’s own weight in its leg chains, and showed why the intuitive remedy—venting more slowly—cannot help: the force is released in

one step when the lip peels, not gradually as the pressure decays. Because attached feet cannot slip, the remedy can be implemented entirely in the foot commands: return each leg’s stored deflection before venting it, hand the load to the remaining legs under mean-invariant, rotation-weighted shares, and the seal breaks with nothing to release. With no force sensing, no new hardware, and one offline calibration, the ratchet’s rupture component falls by 88% and the total by 61%; what remains obeys a simple per-millimeter transfer law whose coefficient is set by load-share allocation, which is the target of our next work.

MULTIMEDIA

[TODO: Attachment video (mandatory for this venue): A-vs-C side-by-side in-place runs + 30 fps slow-motion of a single rupture; source MOVs frozen on the lab machine; ≤ 20 MB/180 s if repurposed for conference submission.]

DATA AVAILABILITY

[TODO: Release bundle: nine + six run logs, trajectories and decomposition reports, 188 verification crops, analysis pipeline (tracking, sync, decomposition, fitting), and the δ -calibration protocol. Decide host (repo/Zenodo) and license.]

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